CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 4 : Python Program for Crypt-Arithmetic Problem

Aim

To develop a Python program that solves the crypt-arithmetic puzzle SEND + MORE = MONEY by assigning a unique digit to each letter.

Algorithm

- List the distinct letters in the puzzle: S, E, N, D, M, O, R, Y.
- Generate every possible assignment of different digits (0–9) to these letters using permutations.
- Reject any assignment in which S or M is 0, since they are leading digits.
- Compute the numeric values of SEND, MORE and MONEY for the assignment.
- Check whether SEND + MORE equals MONEY.
- If it does, print the values of SEND, MORE and MONEY and stop.

Program

```
import itertools letters = "SENDMORY" digits =

range(10) for p in itertools.permutations(digits,

len(letters)):

d = dict(zip(letters, p))     if

d["S"] == 0 or d["M"] == 0:

continue

    send = 1000*d["S"] + 100*d["E"] + 10*d["N"] + d["D"]     more =
1000*d["M"] + 100*d["O"] + 10*d["R"] + d["E"]     money = 10000*d["M"]
+ 1000*d["O"] + 100*d["N"] + 10*d["E"] + d["Y"]

    if send + more == money:
print("SEND =", send)
print("MORE =", more)
print("MONEY =", money)         break
```

Output



RESULT

Thus, the Crypt-Arithmetic Problem was solved successfully using Python.